



1.7.6 Wrap-Up: Cool Down

Complete the statements about today's lesson.

1. Something I learned today ...
2. I didn't know ...
3. One question I still have ...



Unit 1, Lesson H1: Symmetries



Benchmark

MA.912.GR.2.4 Determine symmetries of reflection, symmetries of rotation, and symmetries of translation of a geometric figure.



Learning Targets

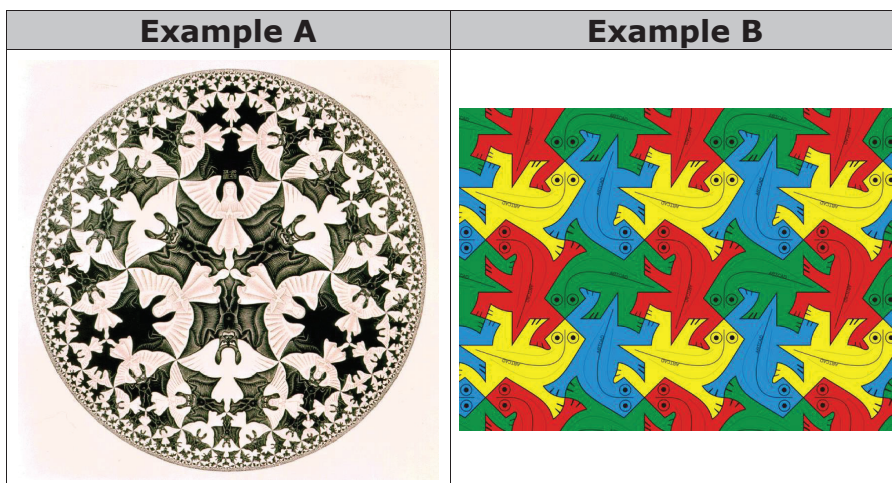
- ☐ I can determine symmetries of reflection.
- ☐ I can determine symmetries of rotation.
- ☐ I can determine symmetries of translation.



1.H1.1 Warm-Up: Artwork of M.C. Escher

M.C. Escher (1898-1972) was a Dutch artist whose work was inspired by mathematics. Though initially trained as an architect, he became one of the most famous and influential graphic artists in the world.

Below are two examples of Escher's work.



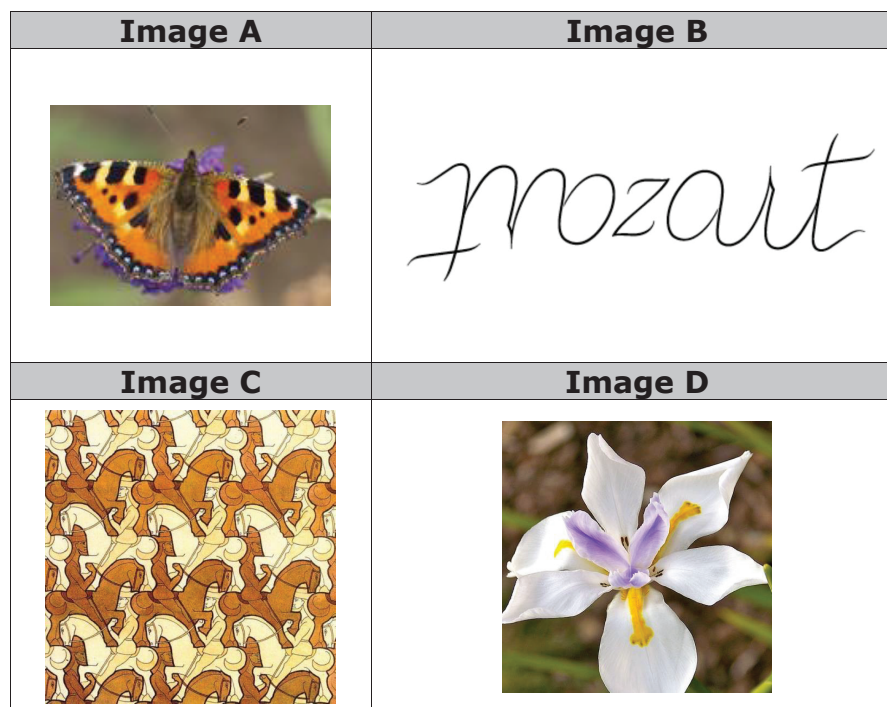
1. Describe any similarities and differences you notice about the two examples.

Similarities	Differences



1.H1.2 Exploration: Symmetry

The four images shown below have symmetry.



1. Discuss with your partner why each image has symmetry and which transformations could be used to show that one part of the image aligns with the other.

2. Complete the table. Some images may align with more than one type of transformation.

Image	Transformations
A	☐ reflection across a line ☐ rotation about a point ☐ translation
B	☐ reflection across a line ☐ rotation about a point ☐ translation
C	☐ reflection across a line ☐ rotation about a point ☐ translation
D	☐ reflection across a line ☐ rotation about a point ☐ translation

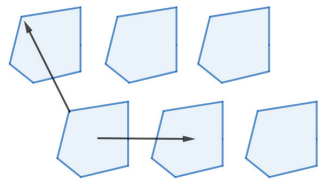
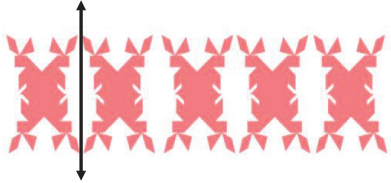
A **tessellation** is an arrangement of shapes closely fitted together in a repeated pattern without gaps or overlaps.

3. Which image is a tessellation?

Translational symmetry is when sliding a pattern is used to create symmetry.

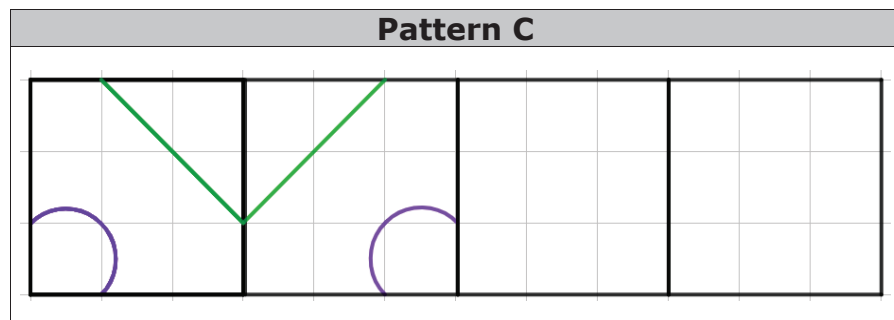
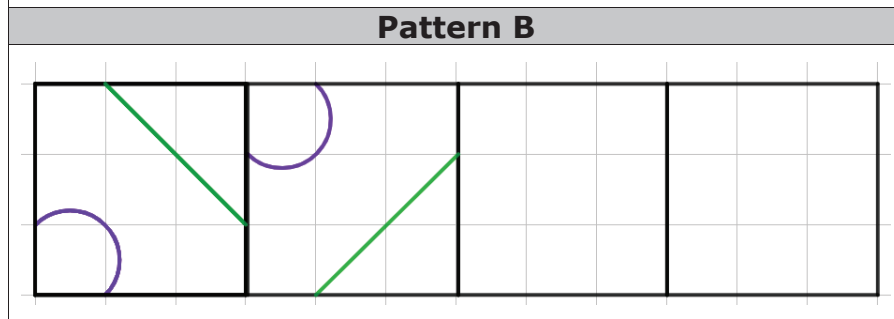
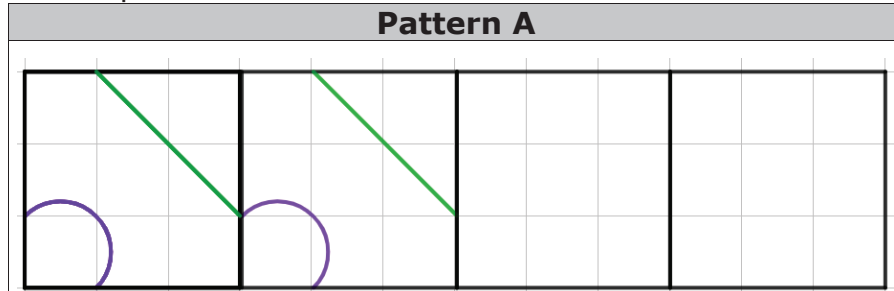
Reflectional symmetry is when a pattern can be reflected to create symmetry.

Rotational symmetry is when a rotation can be used to create symmetry. The table shows examples of each type of symmetry.

Image	Symmetry
	Translational symmetry
	Reflectional symmetry
	Rotational symmetry

4. Micah commented to his partner that he thought the picture for reflectional symmetry could also be translational symmetry. Discuss this with your partner.

5. For each pattern, use the first two squares to complete the pattern.

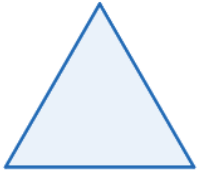


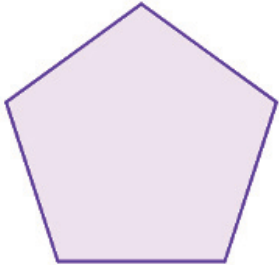
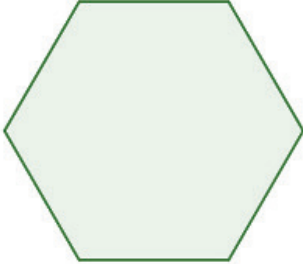
6. Work with your partner to determine the type of symmetry each pattern displays.

Pattern	Type of Symmetry
A	
B	
C	

The order of rotational symmetry of a shape is the number of times it can be rotated around a full circle and still look the same.

7. Complete the chart for each polygon.

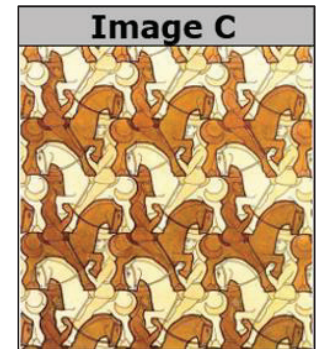
Polygon	Figure	Number of Lines of Symmetry	Order of Rotational Symmetry
Isosceles triangle			
Equilateral triangle			
Square			
Kite			

Polygon	Figure	Number of Lines of Symmetry	Order of Rotational Symmetry
Regular Pentagon			
Regular Hexagon			

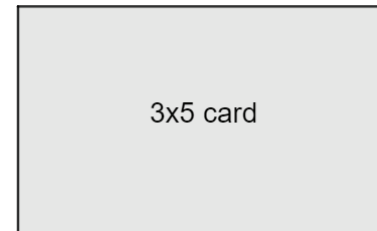


1.H1.3 Activity: Creating a Tessellation

Image C is a tessellation that was created by MC Escher, a famous Dutch artist.



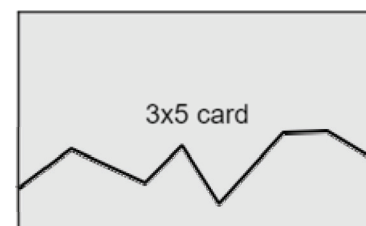
To create a tessellation, you will need a 3x5 index card and tape.



1. Create your own tessellation by choosing one of the following options. The more cuts you make, the more interesting your pattern will become.

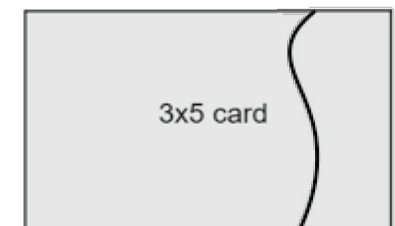
Horizontal Cut

Draw a curve or pattern along a horizontal side of the card.

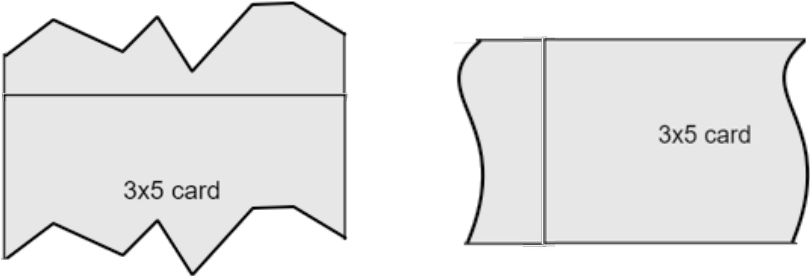


Vertical Cut

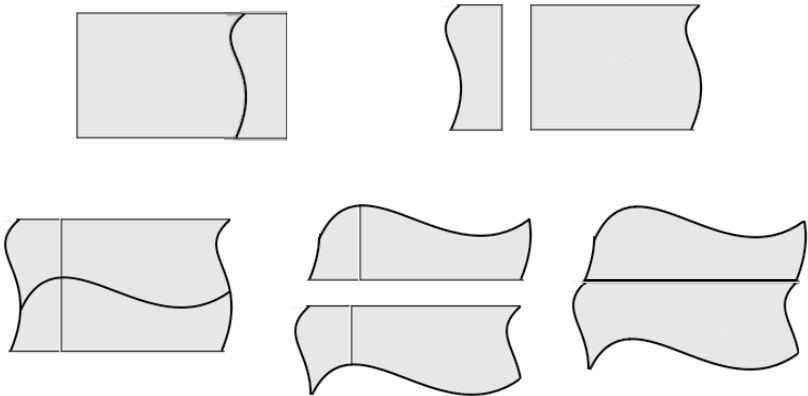
Draw a curve or pattern along a vertical side of the card.



Cut along the curve or pattern and shift the piece to the opposite side. Tape the piece on.



It is also possible to make multiple cuts as shown.



2. Use your card to create a tessellation by filling up a sheet of paper. Then, color your tessellation.



1.H1.4 Wrap-Up: Cool Down

1. Symmetry is often found in nature. Describe the transformations that show the symmetry of each object.

Image	Object	Transformation
	Each petal	
	Each arm of the starfish	

